# STRONGLY CARTESIAN MORPHISMS OF THE QUASI-COHERENT MODULE STACK

THE STACKS PROJECT AUTHORS

*Editorial note.* Statement span is lines 332-337 of sources/stacks-examples-stacks/pinned-original-source.tex; proof span is lines 339-372, the complete author proof (32 body lines).

*Editorial note.* Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

*Editorial note.* extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

*Editorial note.* Text is retained verbatim, including the author's own typos and placeholders.

*Editorial note.* Queue index 3 of queue_examples-stacks.json (pb=1548, stmt 332-337, proof 339-372, halfA).

*Editorial note.* stacks-examples-stacks wave-1 (exs1): fresh adjudication accept (2026-10-14); pool per Chief 09:4x authorization (idle-line throughput over kun's backup note, SMC precedent); examples-stacks.tex 27 labels/20 with proof judged 9 accept / 11 reject; claim_key v2 stacks:examples-stacks.tex:<label>.

*Allowed standard prerequisites (editorial).*

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

*Source reference.* Original lemma 02XK: sources/stacks-examples-stacks/pinned-original-source.tex, source lines 372–372. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 02XL: sources/stacks-examples-stacks/pinned-original-source.tex, source lines 372–372. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 04U2: sources/stacks-examples-stacks/pinned-original-source.tex, source lines 316–321. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

**Lemma 0.1.** *A morphism $(f, \varphi) : (Y, \mathcal{G}) \to (X, \mathcal{F})$ of $\mathcal{QC}oh$ is strongly cartesian if and only if the map $\varphi$ induces an isomorphism $f^*\mathcal{F} \to \mathcal{G}$.*

*Proof.* Let $(X, \mathcal{F}) \in \mathrm{Ob}(\mathcal{QC}oh)$. Let $f : Y \to X$ be a morphism of $(Sch/S)_{fppf}$. Note that there is a canonical $f$-map $c : \mathcal{F} \to f^*\mathcal{F}$ and hence we get a morphism

*Date*: stacks-project:examples-stacks.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

$(f, c) : (Y, f^*\mathcal{F}) \to (X, \mathcal{F})$. We claim that $(f, c)$ is strongly cartesian. Namely, for any object $(Z, \mathcal{H})$ of $\mathcal{QC}oh$ we have

$$\begin{aligned}
\mathrm{Mor}_{\mathcal{QC}oh}((Z, \mathcal{H}), (Y, f^*\mathcal{F})) &= \coprod\nolimits_{g \in \mathrm{Mor}_S(Z,Y)} \mathrm{Mor}_{QCoh(\mathcal{O}_Z)}(g^*f^*\mathcal{F}, \mathcal{H}) \\
&= \coprod\nolimits_{g \in \mathrm{Mor}_S(Z,Y)} \mathrm{Mor}_{QCoh(\mathcal{O}_Z)}((f \circ g)^*\mathcal{F}, \mathcal{H}) \\
&= \mathrm{Mor}_{\mathcal{QC}oh}((Z, \mathcal{H}), (X, \mathcal{F})) \times_{\mathrm{Mor}_S(Z,X)} \mathrm{Mor}_S(Z, Y)
\end{aligned}$$

where we have used Equation (04U2) twice. This proves that the condition of Categories, Definition 02XK holds for $(f, c)$, and hence our claim is true. Now by Categories, Lemma 02XL we see that isomorphisms are strongly cartesian and compositions of strongly cartesian morphisms are strongly cartesian which proves the "if" part of the lemma. For the converse, note that given $(X, \mathcal{F})$ and $f : Y \to X$, if there exists a strongly cartesian morphism lifting $f$ with target $(X, \mathcal{F})$ then it has to be isomorphic to $(f, c)$ (see discussion following Categories, Definition 02XK). Hence the "only if" part of the lemma holds. □